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BATTERY AND ELECTRICAL DEVICE

Abstract

A battery includes a battery cell and a thermal management component. The battery cell includes a first surface and a second surface, and the thermal management component includes a first thermal management component and a second thermal management component. The first thermal management component is arranged on the first surface, and the first thermal management component is configured to adjust the temperature of the battery cell. The second thermal management component is arranged on the second surface, and the second thermal management component is configured to adjust the temperature of the battery cell.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/087898, filed on Apr. 12, 2023, which claims priority to the Chinese patent application with the filing No. 202320031543.4 filed on Jan. 6, 2023, and entitled “BATTERY AND ELECTRICAL DEVICE”, the contents of which are incorporated herein by reference in entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of batteries, and particularly to a battery and an electrical device.

BACKGROUND ART

[0003] With the development of new energy technologies, batteries are widely used in electrical devices. For example, they are used in mobile phones, laptops, electric bicycles, electric cars, electric airplanes, electric ships, electric toy cars, electric toy ships, electric toy airplanes, and electric tools.

[0004] When the battery is operating, the temperature needs to be maintained within an appropriate temperature range. A typical battery usually provides a thermal management component, which is configured to adjust the temperature of the battery. However, the thermal management component is inefficient in adjusting the battery temperature, which in turn reduces the operational efficiency of the battery. Therefore, improving the operational efficiency of the thermal management component of the battery is a technical problem that needs to be urgently addressed.

SUMMARY

[0005] The objective of the present disclosure is to provide a battery and an electrical device, wherein the thermal management component of the battery has the advantage of high operational efficiency.

[0006] On the one hand, an embodiment of the present disclosure provides a battery, which includes a battery cell and a thermal management component. The battery cell includes a first surface and a second surface, and the thermal management component includes a first thermal management component and a second thermal management component. The first thermal management component is arranged on the first surface, and the first thermal management component is configured to adjust the temperature of the battery cell. The second thermal management component is arranged on the second surface, and the second thermal management component is configured to adjust the temperature of the battery cell.

[0007] In the above solution, the first thermal management component and the second thermal management component jointly adjust the temperature of the battery cell, thus improving the temperature adjustment efficiency of the battery cell. Additionally, the first thermal management component and the second thermal management component are arranged on different surfaces of the battery cell, thereby increasing the contact area between the thermal management component and the battery cell. Therefore, the temperature adjustment efficiency of the battery cell is enhanced.

[0008] In some embodiments, the first surface and the second surface are two adjacent surfaces of the battery cell.

[0009] In the above solution, when the first surface and the second surface are adjacent, the contact positions between the first thermal management component, the second thermal management component, and the battery cell are more concentrated, thereby reducing the time required to adjust the temperature of the battery cell, and thus further improving the temperature adjustment efficiency of the battery cell.

[0010] In some embodiments, the first thermal management component is configured to heat or cool the battery cell; and/or the second thermal management component is configured to heat the battery cell.

[0011] In the above solution, when both the first thermal management component and the second thermal management component can be configured to heat the battery cell, the efficiency of heating the battery cell is improved. When the first thermal management component is configured to cool the battery cell, the second thermal management component can be configured to heat the battery cell, thus enabling the battery system to perform corresponding temperature adjustments on the battery cell through different thermal management components when the battery cell is subjected to different ambient temperatures. This allows a wider range of temperatures to be adjusted overall. Moreover, by selectively operating a certain thermal management component, the temperature of battery cells in different regions of the battery can be adjusted. This makes the overall temperature of the battery more uniform and improves the temperature adjustment efficiency of the battery cell.

[0012] In some embodiments, the first thermal management component includes a first accommodating cavity, and the first accommodating cavity is configured to accommodate a heat exchange fluid.

[0013] In the above solution, the first thermal management component, which accommodates the heat exchange fluid, can enable heat to be uniformly distributed through the flow of fluid in the first accommodating cavity, thereby reducing the probability of heat concentration in local regions and also increasing heat transfer efficiency. This improves the heat exchange effect of the first thermal management component.

[0014] In some embodiments, the second thermal management component is an electric heating component.

[0015] In the above solution, the second thermal management component is an electric heating component. On the one hand, the electric heating component has the advantage of rapid temperature increase, which enhances the temperature adjustment efficiency of the battery cell. On the other hand, the electric heating component is powered through a conducting wire, and the conducting wire is easy to arrange, thereby facilitating the mounting and arrangement of the second thermal management component. In some embodiments, the first thermal management component includes an accommodating cavity that houses a heat exchange fluid, the second thermal management component is an electric heating component, and the first thermal management component and the second thermal management component adopt different forms of thermal management arrangements. This can reduce the risk of simultaneous failure of both components leading to uncontrolled thermal management within the battery, thereby improving the overall thermal management reliability of the battery.

[0016] In some embodiments, the battery also includes a first buffering component, and at least a portion of the first buffering component is arranged between the battery cell and the second thermal management component.

[0017] In the above solution, the first buffering component can buffer the pressure exerted by the battery cell on the second thermal management component when the battery cell expands, thereby reducing the risk of damage to the second thermal management component due to pressure and enhancing the safety of the battery.

[0018] In some embodiments, the first buffering component and the second thermal management

component are arranged in a stacked manner.

[0019] In the above solution, the stacked arrangement of the first buffering component and the second thermal management component allows only one surface wall of the second thermal management component being shielded by the first buffering component. This ensures the buffering effect of the first buffering component, and at the same time, a portion of the second thermal management component that is not shielded by the first buffering component can be further configured for heating other battery cells, thereby improving the utilization rate of the second thermal management component.

[0020] In some embodiments, a first housing cavity is formed inside the first buffering component, and the second thermal management component is housed within the first housing cavity.

[0021] In the above solution, by housing the second thermal management component within the first housing cavity, on the one hand, the risk of disengaging the second thermal management component and the first buffering component is reduced. On the other hand, since the battery cell is directly against the second thermal management component, the second thermal management component is pressurized and damaged when the battery cell expands; however, this risk is reduced.

[0022] In some embodiments, the battery cell includes two opposing first surfaces, and the two first surfaces are provided with the first thermal management component.

[0023] In the above solution, by arranging the first thermal management component on both opposing first surfaces of the battery cell, the contact area between the first thermal management component and the battery cell is increased, thereby improving the efficiency of the first thermal management component in adjusting the temperature of the battery cell, and also enhancing the uniformity when regulating the temperature of the battery cell.

[0024] In some embodiments, the battery cell includes two opposing second surfaces, and the two second surfaces are provided with the second thermal management component.

[0025] In the above solution, by arranging the second thermal management component on both opposing second surfaces of the battery cell, the contact area between the second thermal management component and the battery cell is increased, thereby improving the efficiency of the second thermal management component in adjusting the temperature of the battery cell, and also enhancing the uniformity when regulating the temperature of the battery cell.

[0026] In some embodiments, the first surface or the second surface is the largest surface of the battery cell.

[0027] In the above solution, by arranging the first thermal management component or the second thermal management component on the largest surface of the battery cell, the contact area between the first thermal management component or second thermal management component and the battery cell is maximized, thus improving the efficiency of temperature regulation for the battery cell.

[0028] In some embodiments, the battery further includes a box with an opening, wherein the box is configured to accommodate the battery cell. The box includes a bottom wall opposite the opening, and a first thermal management component or a second thermal management component is arranged between the battery cell and the bottom wall.

[0029] In the above solution, by arranging the first thermal management component or the second thermal management component between the battery cell and the bottom wall, the surface of the battery cell shielded by the bottom wall can also be temperature-regulated by the first thermal management component or the second thermal management component. This further increases the contact area between the thermal management component and the battery cell, thereby improving the efficiency of the thermal management component in regulating the temperature of the battery cell.

[0030] In some embodiments, the battery cell further includes a third surface, and the thermal management component further includes a third thermal management component. The third

thermal management component is arranged on the third surface, and the third thermal management component is configured to adjust the temperature of the battery cell.

[0031] In the above solution, by arranging the third thermal management component, the contact area between the thermal management component and the battery cell is further increased, thus improving the efficiency of the thermal management component in adjusting the temperature of the battery cell.

[0032] In some embodiments, the first surface, the second surface, and the third surface are adjacent to each other in pairs.

[0033] In the above solution, when the first surface, the second surface, and the third surface are adjacent to each other in pairs, the contact surfaces between the thermal management component and the battery cell are more concentrated, thereby improving the efficiency of temperature adjustment for the battery cell.

[0034] In some embodiments, the battery further includes a box with an opening, where the box is configured to accommodate the battery cell. The box includes a bottom wall opposite the opening, and a third thermal management component is arranged between the battery cell and the bottom wall.

[0035] In the above solution, by arranging the third thermal management component between the battery cell and the bottom wall, the surface of the battery cell shielded by the bottom wall can also be temperature-regulated by the third thermal management component, thus improving the uniformity of temperature adjustment for the battery cell.

[0036] In some embodiments, the third thermal management component is an electric heating component.

[0037] In the above solution, the third thermal management component is an electric heating component. On the one hand, the electric heating component has the advantage of rapid temperature increase, which enhances the temperature adjustment efficiency of the battery cell. On the other hand, the electric heating component is powered through a conducting wire, and the conducting wire is easy to arrange, thereby facilitating the mounting and arrangement of the third thermal management component. Additionally, in embodiments where the second thermal management component is also an electric heating component, the third thermal management component works in conjunction with the second thermal management component, thus further enhancing the efficiency of temperature adjustment for the battery cell.

[0038] In some embodiments, the battery also includes a second buffering component, and at least a portion of the second buffering component is arranged between the battery cell and the third thermal management component.

[0039] In the above solution, the second buffering component can buffer the pressure exerted by the battery cell on the third thermal management component when the battery cell expands, thereby reducing the risk of damage to the third thermal management component due to pressure and enhancing the safety of the battery.

[0040] In some embodiments, the second buffering component and the third thermal management component are arranged in a stacked manner.

[0041] In the above solution, compared to when the second buffering component is not provided and the battery cell directly compresses the third thermal management component, the second buffering component and the third thermal management component are arranged in a stacked manner. When the battery cell expands in the direction where the third thermal management component is arranged, it can preferentially compress the second buffering component, to reduce the probability of failure or damage to the third thermal management component due to compression. Or, when the third thermal management component is therein arranged with a flow channel for containing heat exchange fluid, the likelihood of the third thermal management component being compressed due to the compression of the flow channel, which further results in poor heat exchange performance, can be reduced.

[0042] In some embodiments, a second housing cavity is formed inside the second buffering component, and the third thermal management component is housed within the second housing cavity.

[0043] In the above solution, by housing the third thermal management component within the second housing cavity, on the one hand, the risk of disengaging the third thermal management component and the second buffering component is reduced. On the other hand, since the battery cell is directly against the third thermal management component, the third thermal management component is pressurized and damaged when the battery cell expands; however, this risk is reduced.

[0044] In some embodiments, the battery cell includes two opposing third surfaces, and the two third surfaces are provided with the third thermal management component.

[0045] In the above solution, by arranging the third thermal management component on both opposing third surfaces of the battery cell, the contact area between the third thermal management component and the battery cell is increased, thereby improving the efficiency of the third thermal management component in adjusting the temperature of the battery cell, and also enhancing the uniformity when regulating the temperature of the battery cell.

[0046] In some embodiments, the third surface is the largest surface of the battery cell.

[0047] In the above solution, when the third surface is the largest surface, the contact area between the third thermal management component and the battery cell is maximized, thus improving the efficiency of the third thermal management component in adjusting the temperature of the battery cell.

[0048] In some embodiments, the battery includes multiple rows of battery cells, and the thermal management component includes multiple first thermal management components, wherein the multiple first thermal management components are arranged along a first direction. A row of battery cells is arranged between two adjacent first thermal management components along the first direction.

[0049] In the above solution, since multiple first thermal management components are arranged in the first direction, each row of battery cells can be housed between two adjacent first thermal management components, thus allowing multiple battery cells to share one first thermal management component. This reduces the number of first thermal management components required for the battery and saves on the manufacturing cost of the battery.

[0050] In some embodiments, in each row of battery cells, a second thermal management component is arranged on both sides of each battery cell along a second direction, and a second thermal management component is arranged between two adjacent battery cells. The first direction is perpendicular to the second direction.

[0051] In the above solution, by arranging the second thermal management components on both sides of each battery cell in each row of battery cells along the second direction, it ensures that the temperature of each battery cell in each row of battery cells can be regulated by the corresponding second thermal management component. A second thermal management component is arranged between two adjacent battery cells for heating, thus allowing the two battery cells to share one second thermal management component. This reduces the number of second thermal management components required and lowers the manufacturing cost of the battery.

[0052] In some embodiments, the thermal management component further includes a third thermal management component. The third thermal management component is arranged along the third direction on at least one side of the battery cell, and the third thermal management component is configured to regulate the temperature of the battery cell. The first direction, the second direction, and the third direction are mutually perpendicular.

[0053] In the above solution, by arranging the third thermal management component on at least one side of the battery cell, it ensures that the temperature of each battery cell in each row of battery cells can be regulated by the corresponding third thermal management component.

[0054] In some embodiments, the battery includes at least one first battery cell, at least one second battery cell, a first control module, and a second control module. The thermal management component arranged on the first battery cell is independently controlled by the first control module, and the thermal management component arranged on the second battery cell is independently controlled by the second control module.

[0055] In the above solution, the first control module independently controls the thermal management component of the first battery cell, and the second control module independently controls the thermal management component of the second battery cell, thus enabling independent temperature regulation for the first battery cell and the second battery cell. Furthermore, different adjustments can be made based on the different environmental temperatures surrounding the first battery cell and the second battery cell, or according to the temperature regulation requirements of each battery cell, thereby improving the rationality of the overall thermal management adjustment of the battery.

[0056] In some embodiments, the battery includes at least one first battery pack, at least one second battery pack, a third control module, and a fourth control module. The first battery pack includes at least one battery cell, and the second battery pack includes at least one battery cell. The thermal management components arranged on the battery cells of the first battery pack are independently controlled by the third control module, and the thermal management components arranged on the battery cells of the second battery pack are independently controlled by the fourth control module.

[0057] In the above solution, the third control module independently controls the first battery pack, and the fourth control module independently controls the second battery pack, thus allowing the third control module and the fourth control module to regulate the temperature of the first battery pack and the second battery pack separately. Furthermore, different adjustments can be made based on the different environmental temperatures or the temperature regulation requirements of the first battery pack and the second battery pack, thereby improving the rationality of the overall thermal management adjustment.

[0058] In some embodiments, each row of battery cells includes multiple battery cells, and a third thermal management component is arranged to correspond to one row of battery cells, with each third thermal management component being configured to regulate the temperature of one row of battery cells.

[0059] In the above solution, each third thermal management component is configured to regulate the temperature of a row of battery cells, which reduces the number of third thermal management components required by the battery and lowers the manufacturing cost of the battery.

[0060] In some embodiments, along the third direction, one side of the battery cell is provided with an electrode terminal, and the side of the battery cell away from the electrode terminal is provided with the third thermal management component.

[0061] In the above solution, by arranging the third thermal management component on the side of the battery cell away from the electrode terminal, the third thermal management component can both regulate the temperature of the battery cell and reduce the risk of a third thermal management component forming an interference with the electrode terminal.

[0062] On the other hand, the present disclosure also provides an electrical device that includes the battery from the embodiments described above.

[0063] The above description is only an overview of the technical solution of the present disclosure. To provide a clearer understanding of the technical solutions of the present disclosure, it can be implemented according to the content of the summary. In order to make the above and other objectives, features, and advantages of the present disclosure more apparent and understandable, specific embodiments of the present disclosure are provided below.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0064] Through reading the detailed description of embodiments below, various other advantages and benefits will become apparent to those skilled in the art. The drawings are provided solely for illustrating some embodiments and are not to be considered as a limitation of the present disclosure. Moreover, in all the drawings, the same reference numerals are used to denote the same components. In the drawings,

[0065] FIG. **1** is a schematic diagram of a structure of a vehicle provided in some embodiments of the present disclosure;

[0066] FIG. **2** is a structural schematic diagram of a battery provided in some embodiments of the present disclosure;

[0067] FIG. **3** is a schematic diagram of a structure of a battery cell provided in some embodiments of the present disclosure;

[0068] FIG. **4** is a schematic diagram of a structure of a first thermal management component provided in some embodiments of the present disclosure;

[0069] FIG. **5** is a schematic diagram of a structure of a second thermal management component provided in some embodiments of the present disclosure;

[0070] FIG. **6** is a schematic diagram of a cooperation between a first buffering component and a battery cell according to some embodiments of the present disclosure (with a second thermal management component sandwiched between two first buffering components, between two battery cells);

[0071] FIG. **7** is a schematic diagram of a cooperation between a first buffering component and a battery cell according to other embodiments of the present disclosure (with a second thermal management component housed within a first housing cavity of a first buffering component);

[0072] FIG. **8** is a schematic diagram of a cooperation between a first buffering component and a second thermal management component according to some embodiments of the present disclosure (with a first housing cavity of a first buffering component being a sealed chamber);

[0073] FIG. **9** is a schematic diagram of a cooperation between a first buffering component and a second thermal management component according to other embodiments of the present disclosure (with a first housing cavity of a first buffering component provided with an opening);

[0074] FIG. **10** is a schematic diagram of a structure of a third thermal management component provided in some embodiments of the present disclosure;

[0075] FIG. **11** is a schematic diagram of a cooperation between a second buffering component and a battery cell according to some embodiments of the present disclosure (with a third thermal management component and a second buffering component being arranged in a stacked manner);

[0076] FIG. **12** is a schematic diagram of a cooperation between a second buffering component and a battery cell according to some embodiments of the present disclosure (with a third thermal management component housed in a second housing cavity of a second buffering component, where a first housing cavity is a sealed chamber);

[0077] FIG. **13** is a schematic diagram of a cooperation between a second buffering component and a third thermal management component according to some embodiments of the present disclosure (with a second housing cavity of a second buffering component provided with an opening);

[0078] FIG. **14** is a schematic diagram of a cooperation between a third thermal management component and a battery cell according to some embodiments of the present disclosure (with both third surfaces of the battery cell being provided with third thermal management components);

[0079] FIG. **15** is a schematic diagram of a cooperation between a first buffering component and a battery cell according to further embodiments of the present disclosure (with one first buffering component and one second thermal management component arranged between two battery cells);

[0080] FIG. **16** is a structural schematic diagram of a battery (showing a first battery pack and a second battery pack) according to some embodiments of the present disclosure; and

[0081] FIG. 17 is a schematic diagram of a cooperation between a third thermal management component and a battery cell according to some embodiments of the present disclosure (with a third thermal management component arranged on a side of a battery cell that faces away from an electrode terminal along a third direction).

[0082] The drawings in the specific embodiments are labeled as follows: [0083] **10**—box; **11**—first part; **12**—second part; **20**—battery cell; **21**—first battery pack; **22**—second battery pack; **201**—first surface; **202**—second surface; **203**—third surface; **204**—electrode terminal; **30**—first thermal management component; **31**—plate body; **32**—first current collector; **321**—medium inlet; **33**—second current collector; **331**—medium outlet; **40**—third thermal management component; **41**—second heating body; **42**—second connecting portion; **50**—second thermal management component; **51**—first heating body; **52**—first connecting portion; **60**—first buffering component; **70**—second buffering component; **80**—fourth thermal management component; **90**—thermal management component; **100**—battery; **200**—controller; **300**—motor; **1000**—vehicle.

DETAILED DESCRIPTION OF EMBODIMENTS

[0084] The following will provide a detailed description of the embodiments of the technical solution of the present disclosure, in conjunction with the accompanying drawings. The following examples are provided solely for a clearer illustration of the technical solutions in the present disclosure. They are presented as examples and should not be used to limit the scope of protection of the present disclosure.

[0085] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as understood by those skilled in the art of the technical field of the present disclosure. The terms used herein are used for the purpose of describing specific embodiments and are not intended to limit the present disclosure. The terms “comprise”, “include”, and any variations thereof as used in the description, claims, and the above-mentioned drawing descriptions of the present disclosure are intended to cover non-exclusive inclusions.

[0086] In the description of the embodiments of the present disclosure, the technical terms “first”, “second”, etc. are used solely for the purpose of distinguishing different objects and should not be construed as indicating or implying relative importance or suggesting a specific quantity, specific order, or hierarchical relationship of the indicated technical features. In the description of the embodiments of the present disclosure, “multiple” means at least two, such as two and three, unless otherwise expressly and specifically limited.

[0087] The term “embodiment” used herein means that specific features, structures, or characteristics described in conjunction with the embodiment can be included in at least one embodiment of the present disclosure. The use of this phrase in various parts of the summary does not necessarily refer to the same embodiment, nor does it imply that these embodiments are mutually exclusive or independent of each other. Those skilled in the art understand explicitly and implicitly that the embodiments described herein can be combined with other embodiments.

[0088] In the description of the embodiments of the present disclosure, the term “multiple” refers to two or more.

[0089] Additionally, in the description of the embodiments of the present disclosure, technical terms indicating orientations or positional relationships, such as “thickness”, “up”, and “down” are based on the orientations or positional relationships shown in the drawings. These terms are used for facilitating the description of the embodiments of the present disclosure and simplifying the description, and should not be construed as indicating or implying that the devices or components referred to must have specific orientations or be constructed and operated in specific orientations. Therefore, they should not be understood as limiting the embodiments of the present disclosure.

[0090] In the description of the embodiments of the present disclosure, unless otherwise expressly specified and limited, technical terms such as “provide”, “connect”, “communicate”, “fix”, and similar terms should be broadly interpreted. For example, they can be fixed connections, detachable connections, or integral connections. They can also be mechanical connections or electrical

connections. They can also be direct connections or indirect connections through intermediate media. Furthermore, they can be internal connections within two components or the interaction between two components. Those of ordinary skill in the art can understand the meanings of the above terms in the embodiments of the present disclosure according to specific situations.

[0091] In the embodiments of the present disclosure, identical reference numerals indicate the same components. For the sake of conciseness, detailed explanations of the same components are omitted in different embodiments. It should be understood that the dimensions such as thickness, length, and width of various components in the embodiments of the present disclosure as shown in the drawings, and the overall dimensions such as thickness, length, and width of the integrated device, are provided as illustrative examples and should not be construed to limit the scope of the present disclosure.

[0092] A battery typically includes one or more battery cells, and the battery cell is an energy storage unit within the battery. Electrical energy is charged into the battery cell and released outward from the battery cell. During operation, the working temperature of a battery cell must be maintained within an appropriate temperature range. If the temperature is too low, the charge and discharge performance of the battery cell decreases, thus reducing the operational efficiency of the battery. If the temperature is too high, it increases the risk of thermal runaway in the battery cell, thus reducing the safety of the battery. Therefore, temperature regulation for the battery mainly focuses on regulating the temperature of the battery cell.

[0093] Commonly, the temperature of the battery cell is regulated through a thermal management component, and the thermal management component can be a water-cooling plate. The water-cooling plate is a plate body with a space formed inside to hold a medium, and the water-cooling plate is applied to the battery cell so that the temperature of the battery cell is managed by the water-cooling plate. If the temperature of the battery cell is too high, the water-cooling plate cools the battery cell; and if the temperature of the battery cell is too low, the water-cooling plate heats the battery cell.

[0094] The inventors noted that in common batteries, the battery cell is typically in contact with the water-cooling plate on only one side, which limits the contact area between the water-cooling plate and the battery cell. This makes it difficult for the water-cooling plate to adjust the temperature of the battery cell in a timely manner when the temperature of the battery cell is too low or too high.

[0095] To address the above issue in the related art, the inventors conducted in-depth research and designed a battery. The battery cell of the battery includes a first surface and a second surface, and the thermal management component includes a first thermal management component and a second thermal management component. The first thermal management component is arranged on the first surface, and the first thermal management component is configured to adjust the temperature of the battery cell. The second thermal management component is arranged on the second surface, and the second thermal management component is configured to adjust the temperature of the battery cell.

[0096] On the one hand, the first thermal management component and the second thermal management component jointly adjust the temperature of the battery cell, thus improving the temperature adjustment efficiency of the battery cell. On the other hand, the first thermal management component and the second thermal management component are arranged on different surfaces of the battery cell, thereby increasing the contact area between the thermal management component and the battery cell. Therefore, the temperature adjustment efficiency of the battery cell is enhanced.

[0097] The embodiments of the present disclosure provide an electrical device that uses batteries as a power source. The electrical device can comprise, but is not limited to, mobile phones, tablets, laptops, electric toys, power tools, electric bicycles, electric cars, boats, spacecraft, and more. The electric toys can include stationary or mobile electric toys, for example, video game consoles, electric car toys, electric boat toys, and electric airplane toys. The spacecraft can include aircraft, rockets, space shuttles, and spacecraft, among others.

[0098] For ease of explanation, the following embodiment will be described using one embodiment of the present disclosure where the electrical device is a vehicle as an example.

[0099] Referring to FIG. 1, FIG. 1 is a schematic diagram of a structure of a vehicle **1000** provided in some embodiments of the present disclosure. The vehicle **1000** can be a gasoline-powered car, a natural gas car, or a new energy vehicle. The new energy vehicles can be pure electric cars, hybrid cars, extended-range cars, etc. A battery **100** is provided inside the vehicle **1000**, wherein the battery **100** can be arranged at the bottom, front, or rear of the vehicle **1000**. The battery **100** is configured to provide power for the vehicle **1000**. For instance, the battery **100** can serve as the operational power source for the vehicle **1000**. Additionally, the vehicle **1000** also comprises a controller **200** and a motor **300**. The controller **200** is configured to manage the supply of power from the battery **100** to the motor **300**, e.g., for working power requirements for starting, navigating, and driving the vehicle **1000**.

[0100] In some embodiments of the present disclosure, the battery **100** serves not only as the operational power source for the vehicle **1000** but also as the driving power source for the vehicle **1000**. It can replace or partially substitute for conventional fuels like gasoline or natural gas to provide driving propulsion for the vehicle **1000**.

[0101] According to some embodiments of the present disclosure, referring to FIG. 2 and FIG. 3, FIG. 2 is a structural schematic diagram of a battery **100** provided in some embodiments of the present disclosure, and FIG. 3 is a structural schematic diagram of a battery cell **20** provided in some embodiments of the present disclosure. The embodiment of the present disclosure provides a battery **100**, which includes a battery cell **20** and a thermal management component **90**. The battery cell **20** includes a first surface **201** and a second surface **202**, and the thermal management component **90** includes a first thermal management component **30** and a second thermal management component **50**. The first thermal management component **30** is arranged on the first surface **201**, and the first thermal management component **30** is configured to adjust the temperature of the battery cell **20**. The second thermal management component **50** is arranged on the second surface **202**, and the second thermal management component **50** is configured to adjust the temperature of the battery cell **20**.

[0102] The battery cell **20** refers to the smallest unit comprising the battery **100**. The battery cell **20** can be a secondary battery or a primary battery; it can also be a lithium-sulfur battery, sodium-ion battery, magnesium-ion battery, or other types, but is not limited to these.

[0103] In the battery **100**, one or multiple battery cells **20** can be provided. The multiple battery cells **20** can be connected in series, parallel, or hybrid. A hybrid connection refers to having both series and parallel connections among multiple battery cells **20**. The battery **100** can also comprise other structures. For example, the battery **100** can also comprise a busbar component, which is configured for achieving electrical connections between the multiple battery cells **20**.

[0104] When the battery **100** includes multiple battery cells **20**, the multiple battery cells **20** can be arranged in one or more rows, or stacked in multiple layers, with each layer of battery cells **20** arranged in one or more rows.

[0105] The battery **100** can be provided with a container configured for housing the battery cells **20**, or alternatively, the battery **100** cannot include a container for housing the battery cells **20**. The electrical device can be provided with a mounting chamber for the battery cells **20**.

[0106] The first surface **201** and the second surface **202** are each a certain outer surface of the battery cell **20**. The first surface **201** and the second surface **202** can be adjacent to each other or non-adjacent.

[0107] The first thermal management component **30** can be a component with a warming function, a component with a cooling function, or a component with both a warming and a cooling function.

[0108] The first thermal management component **30** can directly abut the first surface **201** of the battery cell **20** for heat exchange, or it can exchange heat with the first surface **201** of the battery cell **20** through a conduction element. For example, the first surface **201** can be coated with a

thermal conductive adhesive layer, and the first thermal management component **30** can adhere to the first surface **201** through the thermal conductive adhesive layer.

[0109] One or multiple first thermal management components **30** can be provided, and multiple first thermal management components **30** can be arranged only on one first surface **201** of the battery cell **20**, or the battery cell **20** can also include multiple first surfaces **201**, where multiple first thermal management components **30** correspond one-to-one with multiple first surfaces **201**.

[0110] When multiple first thermal management components **30** are arranged on one first surface **201** of the battery cell **20**, the multiple first thermal management components **30** can be arranged in a stacked manner, that is, in the multiple first thermal management components **30**, only one first thermal management component **30** is in contact with the battery cell **20**. The multiple first thermal management components **30** can also be provided side-by-side. That is, the multiple first thermal management components **30** are all in contact with the battery cell **20**.

[0111] The second thermal management component **50** can be a component with a warming function, a component with a cooling function, or a component with both a warming and a cooling function.

[0112] The second thermal management component **50** can directly abut the second surface **202** of the battery cell **20** for heat exchange, or it can exchange heat with the first surface **201** of the battery cell **20** through a conduction element. For example, the second surface **202** can be coated and provided with a thermal conductive adhesive layer, and the first thermal management component **30** can adhere to the second surface **202** through the thermal conductive adhesive layer.

[0113] One or multiple second thermal management components **50** can be provided, and multiple second thermal management components **50** can be arranged only on one second surface **202** of the battery cell **20**, or the battery cell **20** can also include multiple second surfaces **202**, where multiple second thermal management components **50** correspond one-to-one with multiple second surfaces **202**.

[0114] When multiple second thermal management components **50** are arranged on one second surface **202** of the battery cell **20**, the multiple second thermal management components **50** can be arranged in a stacked manner, that is, in the multiple second thermal management components **50**, only one second thermal management component **50** is in contact with the battery cell **20**. The multiple second thermal management components **50** can also be provided side-by-side. That is, the multiple second thermal management components **50** are all in contact with the battery cell **20**.

[0115] The first thermal management component **30** and the second thermal management component **50** can be of the same type of thermal management component. For example, both the first thermal management component **30** and the second thermal management component **50** can be water-cooling plates. The first thermal management component **30** and the second thermal management component **50** can also be different types of thermal management components. For example, the first thermal management component **30** can be a water-cooling plate, and the second thermal management component **50** can be an electric heating component. In the embodiment where multiple first thermal management components **30** and second thermal management components **50** are provided, the first thermal management component **30** and the second thermal management component **50** can be partially of the same type of thermal management component, and partially of different types of thermal management components. For example, the first thermal management component **30** can include water cooling components and electric heating components, and the second thermal management component **50** can include only electric heating components.

[0116] In the battery **100** of the embodiments of the present disclosure, firstly, by jointly regulating the temperature of the battery cell **20** through the first thermal management component **30** and the second thermal management component **50**, the temperature adjustment efficiency of the battery cell **20** is improved. Secondly, the first thermal management component **30** and the second thermal management component **50** are arranged on different surfaces of the battery cell **20**, which

increases the contact area between the thermal management component **90** and the battery cell **20**, and thus improves the temperature adjustment efficiency of the battery cell **20**.

[0117] According to some embodiments of the present disclosure, referring to FIG. 3, the first surface **201** and the second surface **202** are two adjacent surfaces of the battery cell **20**.

[0118] When the first surface **201** and the second surface **202** are adjacent, the first surface **201** and the second surface **202** can be directly connected, or a transition surface can be provided between the first surface **201** and the second surface **202**, such as a transition curved surface or a transition chamfered surface.

[0119] When the first surface **201** and the second surface **202** are adjacent, the contact positions between the first thermal management component **30**, the second thermal management component **50**, and the battery cell **20** are more concentrated, thereby reducing the time required to adjust the temperature of the battery cell **20**, and thus further improving the temperature adjustment efficiency of the battery cell **20**.

[0120] According to some embodiments of the present disclosure, the first thermal management component **30** is configured to heat or cool the battery cell **20**; and/or the second thermal management component **50** is configured to heat the battery cell **20**.

[0121] When the first thermal management component **30** is configured to heat the battery cell **20**, the second thermal management component **50** can be configured to heat the battery cell **20**.

[0122] When the first thermal management component **30** is configured to cool the battery cell **20**, the second thermal management component **50** can also be configured to heat the battery cell **20**.

[0123] It can be understood that the first thermal management component **30** is a thermal management component capable of both heating the battery cell **20** and cooling the battery cell **20**, and the second thermal management component **50** is a heating component capable of heating the battery cell **20**.

[0124] When both the first thermal management component **30** and the second thermal management component **50** are configured to heat the battery cell **20**, the efficiency of heating the battery cell **20** is improved. When the first thermal management component **30** is configured to cool the battery cell **20**, the second thermal management component **50** can be configured to heat the battery cell **20**, thus enabling the battery thermal management system to perform corresponding temperature adjustments through different thermal management components when the battery cell **20** is subjected to different ambient temperatures. This allows for a wider overall adjustable temperature range and greater maneuverability of the battery thermal management system. Moreover, the temperature of the battery cell **20** can also be adjusted locally by different thermal management components, making the overall temperature of the battery cell **20** more uniform and improving the effect of the temperature regulation of the battery cell **20**.

[0125] According to some embodiments of the present disclosure, the first thermal management component **30** is configured to heat or cool the battery cell **20**; and/or the second thermal management component **50** is configured to cool the battery cell **20**.

[0126] When the first thermal management component **30** is configured to heat the battery cell **20**, the second thermal management component **50** can be configured to cool the battery cell **20**.

[0127] When the first thermal management component **30** is configured to cool the battery cell **20**, the second thermal management component **50** can also be configured to cool the battery cell **20**.

[0128] When both the first thermal management component **30** and the second thermal management component **50** are configured to cool the battery cell **20**, the efficiency of cooling the battery cell **20** is improved. When the first thermal management component **30** is configured to heat the battery cell **20**, the second thermal management component **50** is configured to cool the battery cell **20**. This facilitates the adjustment of the temperature of the localized temperature of the battery cell **20**, thus making the overall temperature of the battery cell **20** more uniform and improving the effect of the temperature regulation of the battery cell **20**.

[0129] According to some embodiments of the present disclosure, referring to FIG. 4, FIG. 4 is a

schematic diagram of a structure of a first thermal management component **30** provided in some embodiments of the present disclosure. The first thermal management component **30** includes a first accommodating cavity, and the first accommodating cavity is configured to accommodate a heat exchange fluid.

[0130] The first thermal management component **30** can include a plate body **31**, a first current collector **32**, and a second current collector **33**, with multiple flow channels formed inside the plate body **31** to form a first accommodating cavity. The inlets of the multiple flow channels are in communication with the first current collector **32**, and the outlets of the multiple flow channels are in communication with the second current collector **33**. The first current collector **32** is provided with a medium inlet **321**, and the second current collector **33** is provided with a medium outlet **331**.

[0131] The operation process of the first thermal management component **30** can be as follows. A fluid medium can enter the first current collector **32** through the medium inlet **321**, then flow into each of the flow channels, and the fluid medium discharged from the flow channels is collected in the second current collector **33** and discharged through the medium outlet **331**.

[0132] The plate body **31** can directly abut the battery cell **20** for heat exchange, or it can perform heat exchange with the battery cell **20** through a conduction element. For example, the surface wall of the battery cell **20** can be coated with an adhesive layer, and the plate body **31** can be bonded to the battery cell **20** through the adhesive layer.

[0133] The plate body **31** can be a harmonica-tube-type plate body **31**.

[0134] The first thermal management component **30**, which accommodates the heat exchange fluid, can enable heat to be uniformly distributed through the flow of fluid in the accommodating cavity, thereby reducing the probability of heat concentration in local regions and also increasing heat transfer efficiency. This improves the heat exchange effect of the first thermal management component **30**.

[0135] According to some embodiments of the present disclosure, referring to FIG. 5, FIG. 5 is a schematic diagram of a structure of a second thermal management component **50** provided in some embodiments of the present disclosure. The second thermal management component **50** is an electric heating component.

[0136] The electric heating component is a component that converts electrical energy into thermal energy. For example, the electric heating component can include a resistive heating element. When current flows through the resistive heating element, the resistive heating element generates heat to warm the battery cell **20**.

[0137] The second thermal management component **50** can include a first heating body **51** and multiple first connection portions **52**. The multiple first connection portions **52** are electrically connected to the first heating body **51**. The first heating body **51** is configured to adjust the temperature of the battery cell **20**, and the first connection portions **52** are configured to connect the conducting wires. For example, two first connection portions **52** can be provided, one for connecting the positive conducting wire and the other for connecting the negative conducting wire.

[0138] The second thermal management component **50** is an electric heating component. On the one hand, the electric heating component has the advantage of rapid temperature increase, which enhances the temperature adjustment efficiency of the battery cell **20**. On the other hand, the electric heating component is powered through a conducting wire, and the conducting wire is easy to arrange, thereby facilitating the mounting and arrangement of the second thermal management component **50**.

[0139] In some embodiments, the first thermal management component **30** includes an accommodating cavity that houses a heat exchange fluid, the second thermal management component **50** is an electric heating component, and the first thermal management component **30** and the second thermal management component **50** adopt different forms of thermal management arrangements. This can reduce the risk of simultaneous failure of both components leading to uncontrolled thermal management within the battery **100**, thereby improving the overall thermal

management reliability of the battery **100**.

[0140] According to some embodiments of the present disclosure, the second thermal management component **50** is a PTC heater and a heating film.

[0141] The PTC (positive temperature coefficient) heater includes a housing and a heating unit, the heating unit is housed in the housing. The housing can directly abut the battery cell **20** for heat exchange, or it can perform heat exchange with the battery cell **20** through a conduction element. For example, the surface wall of the battery cell **20** can be coated with an adhesive layer, and the housing can be bonded to the battery cell **20** through the adhesive layer.

[0142] The heating film is a flexible heating thin plate structure.

[0143] The PTC heater has the advantage of a long service life, thereby reducing the usage cost of the battery **100**. The heating film has a soft, deformable structure that can tightly adhere to the battery cell **20**, thus increasing the contact area between the second thermal management component **50** and the battery cell **20**. Therefore, this improves the heating efficiency of the battery cell **20**.

[0144] The specific structure and working principle of the PTC heater and heating film are well known to those skilled in the art, and will not be further elaborated herein.

[0145] According to some embodiments of the present disclosure, the second thermal management component **50** can include a second accommodating cavity, and the second accommodating cavity is configured to house a heat exchange fluid.

[0146] According to some embodiments of the present disclosure, referring to FIG. 6 and FIG. 7, FIG. 6 is a schematic diagram of a cooperation between a first buffering component **60** and a battery cell **20** according to some embodiments of the present disclosure (with a second thermal management component **50** sandwiched between two first buffering components **60**, between two battery cells **20**); FIG. 7 is a schematic diagram of a cooperation between a first buffering component **60** and a battery cell **20** according to other embodiments of the present disclosure (with a second thermal management component **50** housed within a first housing cavity of a first buffering component **60**). The battery **100** also includes a first buffering component **60**, and at least a portion of the first buffering component **60** is arranged between the battery cell **20** and the second thermal management component **50**.

[0147] The first buffering component **60** is a component capable of deformation under pressure, and the first buffering component **60** can also be made of an insulating material, thus ensuring electrical insulation between the second thermal management component **50** and the battery cell **20**. This improves the safety of the battery **100**.

[0148] In the embodiment, the first buffering component **60** can be made of rubber or soft polyvinyl chloride.

[0149] The first buffering component **60** can be a sheet-like structure, arranged between the first surface **201** of the battery cell **20** and the second thermal management component **50**.

[0150] The first buffering component **60** can also wrap around the battery cell **20**, and partially be positioned between the first surface **201** of the battery cell **20** and the second thermal management component **50**.

[0151] The first buffering component **60** can be arranged on the side of the second thermal management component **50** which is closer to the battery cell **20**, or on the side of the second thermal management component **50** which is farther from the battery cell **20**.

[0152] When the battery cell **20** is provided on both sides of the second thermal management component **50**, the first buffering component **60** can be arranged on both sides of the second thermal management component **50**, or it can only be arranged on one side.

[0153] The first buffering component **60** can buffer the pressure exerted by the battery cell **20** on the second thermal management component **50** when the battery cell **20** expands, thereby reducing the risk of damage to the second thermal management component **50** due to pressure and enhancing the safety of the battery **100**.

[0154] According to some embodiments of the present disclosure, with reference to FIG. 6, the first buffering component **60** and the second thermal management component **50** are arranged in a stacked manner.

[0155] The first buffering component **60** can be affixed to the surface wall of the second thermal management component **50** facing the battery cell **20**.

[0156] The first buffering component **60** can abut the second thermal management component **50**, or it can be bonded to the second thermal management component **50**.

[0157] The first buffering component **60** and the second thermal management component **50** are arranged in a stacked manner, thus allowing the battery cell **20** to preferentially compress the first buffering component **60** when the battery cell **20** expands in the direction where the second thermal management component **50** is arranged. This reduces the probability of failure or damage to the second thermal management component **50** due to compression. Alternatively, when the second thermal management component **50** is therein arranged with a flow channel for containing heat exchange fluid, the likelihood of the second thermal management component **50** being compressed due to the compression of the flow channel, which further results in poor heat exchange performance, can be reduced.

[0158] According to some embodiments of the present disclosure, referring to FIG. 8 and FIG. 9, FIG. 8 is a schematic diagram of a cooperation between a first buffering component **60** and a second thermal management component **50** according to some embodiments of the present disclosure (with a first housing cavity of a first buffering component **60** being a sealed chamber); FIG. 9 is a schematic diagram of a cooperation between a first buffering component **60** and a second thermal management component **50** according to other embodiments of the present disclosure (with a first housing cavity of a first buffering component **60** provided with an opening). A first housing cavity is formed inside the first buffering component **60**, and the second thermal management component **50** is housed within the first housing cavity.

[0159] The first housing cavity can be a closed chamber, and the second thermal management component **50** can be completely housed in the first housing cavity.

[0160] The first housing cavity can also be a chamber arranged with an opening, and the second thermal management component **50** can enter the first housing cavity through the opening of the first housing cavity.

[0161] By housing the second thermal management component **50** within the first housing cavity, on the one hand, the risk of disengaging the second thermal management component **50** and the first buffering component **60** is reduced. On the other hand, since the battery cell **20** is directly against the second thermal management component **50**, the second thermal management component **50** is pressurized and damaged when the battery cell **20** expands; however, this risk is reduced.

[0162] According to some embodiments of the present disclosure, when the second thermal management component **50** is a heating film, the second thermal management component **50** can be bent to form an internally hollow cylindrical structure, and the first buffering component **60** can be inserted within the second thermal management component **50**.

[0163] According to some embodiments of the present disclosure, with reference to FIG. 2 and FIG. 3, the battery cell **20** includes two opposing first surfaces **201**, and the two first surfaces **201** are provided with the first thermal management component **30**.

[0164] The two first surfaces **201** can be in a facing state, meaning the normal of the two first surfaces **201** are collinear. For example, the battery cell **20** can be rectangular in shape, wherein the two first surfaces **201** can be the two surfaces opposite each other in the length direction of the battery cell **20**; and it can also be two surfaces opposite each other in the width direction, and can also be two surfaces opposite each other in the thickness direction.

[0165] The two first surfaces **201** can also be in a non-facing state, where the angle between the normal of the two first surfaces **201** is less than 90°.

[0166] Each of the two first surfaces **201** can be provided with one first thermal management component **30**, or they can each be provided with multiple first thermal management components **30**.

[0167] By arranging the first thermal management component **30** on both opposing first surfaces **201** of the battery cell **20**, the contact area between the first thermal management component **30** and the battery cell **20** is increased, thereby improving the efficiency of the first thermal management component **30** in adjusting the temperature of the battery cell **20**, and also enhancing the uniformity when regulating the temperature of the battery cell **20**.

[0168] According to some embodiments of the present disclosure, with reference to FIG. 2 and FIG. 3, the battery cell **20** includes two opposing second surfaces **202**, and the two second surfaces **202** are provided with the second thermal management component **50**.

[0169] The two second surfaces **202** can be in a facing state, meaning the normal of the two second surfaces **202** are collinear. For example, the battery cell **20** can be rectangular in shape, wherein the two second surfaces **202** can be the two surfaces opposite each other in the length direction of the battery cell **20**; and it can also be two surfaces opposite each other in the width direction, and can also be two surfaces opposite each other in the thickness direction.

[0170] The two second surfaces **202** can also be in a non-facing state, where the angle between the normal of the two second surfaces **202** can be less than 90°.

[0171] Each of the two second surfaces **202** can be provided with one second thermal management component **50**, or they can each be provided with multiple second thermal management components **50**.

[0172] By arranging the second thermal management component **50** on both opposing second surfaces **202** of the battery cell **20**, the contact area between the second thermal management component **50** and the battery cell **20** is increased, thereby improving the efficiency of the second thermal management component **50** in adjusting the temperature of the battery cell **20**, and also enhancing the uniformity when regulating the temperature of the battery cell **20**.

[0173] According to some embodiments of the present disclosure, the first surface **201** or the second surface **202** is the largest surface of the battery cell **20**.

[0174] In the embodiment, the first surface **201** can be the largest surface of the battery cell **20**, or the second surface **202** can be the largest surface of the battery cell **20**. Moreover, the first surface **201** and the second surface **202** can have equal areas and are both the largest surfaces among the multiple surfaces of the battery cell **20**.

[0175] The largest surface of the battery cell **20** can be one or more. For example, if the battery cell **20** is cylindrical, the largest surface of the battery cell **20** can be the outer peripheral surface of the battery cell **20**, in which case the number of the largest surfaces of the battery cell **20** can be one. Further, the largest surfaces of the battery cell **20** can be the two circular end surfaces of the battery cell **20**, in which case the number of the largest surfaces of the battery cell **20** can be two. If the battery cell **20** is a rectangular prism, the number of the largest surfaces of the battery cell **20** can be two or four. If the battery cell **20** is a cube, all surfaces of the battery cell **20** are the largest surfaces.

[0176] If the battery cell **20** is a soft pack battery cell **20**, the surface of the battery cell **20** can be determined by the projected area of the surface on a reference plane, and the reference plane can be a plane facing the surface. For example, if the battery cell **20** is a soft-packed rectangular prism battery cell **20**, the reference planes can include three: a first reference plane perpendicular to the length direction of the battery cell **20**, a second reference plane perpendicular to the width direction of the battery cell **20**, and a third reference plane perpendicular to the thickness direction of the battery cell **20**. The projections of the two surfaces of the battery cell **20** in the length direction on the first reference plane can be approximated as the two surfaces in the length direction of the battery cell **20**, and similarly for the two surfaces in the width direction of the battery cell **20** and the two surfaces of the thickness direction of the battery cell **20**, thus determining the largest

surface of the battery cell **20**.

[0177] By arranging the first thermal management component **30** or the second thermal management component **50** on the largest surface of the battery cell **20**, the contact area between the first thermal management component **30** or second thermal management component **50** and the battery cell **20** is maximized, thus improving the efficiency of temperature regulation for the battery cell **20**.

[0178] In embodiments where the first thermal management component **30** is a water-cooling plate, since the water-cooling plate can both heat the battery cell **20** and cool the battery cell **20**, arranging the first thermal management component **30** on the largest surface of the battery cell **20** can enhance the efficiency of the first thermal management component **30** when heating the battery **100** and when cooling the battery cell **20**.

[0179] According to some embodiments of the present disclosure, the battery **100** further includes a box **10** with an opening, where the box **10** is configured to accommodate the battery cell **20**. The box **10** includes a bottom wall opposite the opening, and a first thermal management component **30** or a second thermal management component **50** is arranged between the battery cell **20** and the bottom wall.

[0180] The box **10** can comprise a first part **11** and a second part **12**, wherein the first part **11** and the second part **12** are fitted together. The first part **11** and the second part **12** together define the holding space configured to house the battery cell **20**. The second part **12** can be a hollow structure open at one end, and the first part **11** can be a plate-like structure. The first part **11** is capped over an opening side of the second part **12** so that the first part **11** and the second part **12** together define a holding space. Additionally, both the first part **11** and the second part **12** can be hollow structures open at one end, and an opening side of the first part **11** is capped over an opening side of the second part **12**. Of course, the box **10** formed by the first part **11** and the second part **12** can be of a variety of shapes, e.g., cylindrical, rectangular, etc.

[0181] The box **10** can be made of a rigid material or be a flexible film that wraps around the module formed by the battery cell **20**.

[0182] The bottom wall can be any of the walls of the box **10** that provides support to the battery cell **20**.

[0183] One or multiple first thermal management components **30** or the second thermal management components **50** can be arranged between the battery cell **20** and the bottom wall.

[0184] The projection of the battery cell **20** on the bottom wall can lie within the range of the first thermal management component **30** or the second thermal management component **50**.

[0185] By arranging the first thermal management component **30** or the second thermal management component **50** between the battery cell **20** and the bottom wall, the surface of the battery cell **20** that is obscured by the bottom wall can also be temperature-adjusted by the first thermal management component **30** or the second thermal management component **50**, thus further increasing the contact area between the thermal management component and the battery cell **20**, and enhancing the efficiency of temperature regulation components in regulating the temperature of the battery cell **20**.

[0186] In the embodiment where the first thermal management component **30** is a water-cooling plate, arranging the first thermal management component **30** between the battery cell **20** and the bottom wall facilitates the placement of piping connected to the water-cooling plate.

[0187] According to some embodiments of the present disclosure, referring to FIG. **10** and FIG. **11**, FIG. **10** is a schematic diagram of a structure of a third thermal management component **40** provided in some embodiments of the present disclosure; and FIG. **11** is a schematic diagram of a cooperation between a second buffering component **70** and a battery cell **20** according to some embodiments of the present disclosure (with a third thermal management component **40** and a second buffering component **70** being arranged in a stacked manner); The battery cell **20** further includes a third surface **203**, and the thermal management component **90** further includes a third

thermal management component **40**. The third thermal management component **40** is arranged on the third surface **203**, and the third thermal management component **40** is configured to adjust the temperature of the battery cell **20**.

[0188] The first surface **201**, the second surface, **202**, and the third surface **203** are adjacent to each other in pairs. Any two of the three surfaces can be adjacent, with the remaining surface not adjacent to the two adjacent surfaces. Or, all three surfaces can be non-adjacent.

[0189] The third thermal management component **40** can be a component with a warming function, a component with a cooling function, or a component with both a warming and a cooling function.

[0190] The third thermal management component **40** can directly abut the third surface **203** of the battery cell **20** for heat exchange, or it can exchange heat with the third surface **203** of the battery cell **20** through a conduction element. For example, the third surface **203** of the battery cell **20** can be coated and provided with a thermal conductive adhesive layer, and the first thermal management component **30** can adhere to the battery cell **20** through the thermal conductive adhesive layer.

[0191] One or multiple third thermal management components **40** can be provided, and multiple third thermal management components **40** can be arranged only on the first one of the third surfaces **203** of the battery cell **20**, or the battery cell **20** can also include multiple third surfaces **203**, where multiple third thermal management components **40** correspond one-to-one with multiple third surfaces **203**.

[0192] When multiple third thermal management components **40** are arranged on one third surface **203** of the battery cell **20**, the multiple third thermal management components **40** can be arranged in a stacked manner, that is, in the multiple first thermal management components **30**, only one third thermal management component **40** is in contact with the battery cell **20**. The multiple third thermal management components **40** can also be provided side-by-side. That is, the multiple third thermal management components **40** are all in contact with the battery cell **20**.

[0193] The first thermal management component **30**, the second thermal management component **50**, and the third thermal management component **40** can all be of the same type of thermal management component. The third thermal management component **40** can also be a thermal management component of the same type as only the second thermal management component **50**, or a thermal management component of the same type as only as the first thermal management component **30**.

[0194] By arranging the third thermal management component **40**, the contact area between the thermal management component **90** and the battery cell **20** is further increased, thus improving the efficiency of the thermal management component **90** in adjusting the temperature of the battery cell **20**.

[0195] According to some embodiments of the present disclosure, referring to FIG. 3, the first surface **201**, the second surface **202**, and the third surface **203** are adjacent in pairs.

[0196] When the first surface **201**, the second surface **202**, and the third surface **203** are adjacent in pairs, the first surface **201**, the second surface **202**, and the third surface **203** can intersect at a vertex or at a face. For example, the first surface **201**, the second surface **202**, and the third surface **203** can intersect at a curved surface.

[0197] Any two surfaces among the first surface **201**, the second surface **202**, and the third surface **203** can be directly connected, or the two surfaces can have a transitional surface, such as a transitional curved surface or transitional chamfered surface.

[0198] When the first surface **201**, the second surface **202**, and the third surface **203** are adjacent to each other in pairs, the contact surfaces between the thermal management component **90** and the battery cell **20** are more concentrated, thereby improving the efficiency of temperature adjustment for the battery cell **20**.

[0199] According to some embodiments of the present disclosure, with reference to FIG. 2, the battery **100** further includes a box **10** with an opening, where the box **10** is configured to accommodate the battery cell **20**. The box **10** includes a bottom wall opposite the opening, and a

third thermal management component **40** is arranged between the battery cell **20** and the bottom wall.

[0200] In the embodiment, the third surface **203** can be the surface of the battery cell **20** facing the bottom wall.

[0201] On the side of the battery cell **20** that faces away from the bottom wall, a third thermal management component **40** can also be provided.

[0202] One third thermal management component **40** can be arranged between the battery cell **20** and the bottom wall, or multiple third thermal management components **40** can be arranged.

[0203] By arranging the third thermal management component **40** between the battery cell **20** and the bottom wall, the surface of the battery cell **20** shielded by the bottom wall can also be temperature-regulated by the third thermal management component **40**, thus improving the uniformity of temperature adjustment for the battery cell **20**.

[0204] According to some embodiments of the present disclosure, the third thermal management component **40** is an electric heating component.

[0205] The third thermal management component **40** can include a second heating body **41** and multiple second connection portions **42**. The multiple second connection portions **42** are electrically connected to the second heating body **41**. The second heating body **41** is configured to adjust the temperature of the battery cell **20**, and the second connection portions **42** are configured to connect the conducting wires. For example, two second connection portions **42** can be provided, one for connecting the positive conducting wire and the other for connecting the negative conducting wire.

[0206] The third thermal management component **40** is an electric heating component. On the one hand, the electric heating component has the advantage of rapid temperature increase, which enhances the temperature adjustment efficiency of the battery cell **20**. On the other hand, the electric heating component is powered through a conducting wire, and the conducting wire is easy to arrange, thereby facilitating the mounting and arrangement of the third thermal management component **40**. Additionally, in embodiments where the second thermal management component **50** is also an electric heating component, the third thermal management component **40** works in conjunction with the second thermal management component **50**, thus further enhancing the efficiency of temperature adjustment for the battery cell **20**.

[0207] According to some embodiments of the present disclosure, the third thermal management component **40** is a PTC heater and a heating film.

[0208] The PTC heater has the advantage of a long service life, The selection of the third thermal management component **40** as a PTC heater reduces the usage cost of the battery **100**. The heating film has a soft, deformable structure that can tightly adhere to the battery cell **20**, thus increasing the contact area between the third thermal management component **40** and the battery cell **20**.

Therefore, this improves the heating efficiency of the battery cell **20**. In the embodiment where the third thermal management component **40** is arranged between the bottom wall and the battery cell **20**, the third thermal management component **40** is a heating film, which can reduce the risk of the third thermal management component **40** being damaged due to pressure.

[0209] The third thermal management component **40** can include a third accommodating cavity, and the third accommodating cavity is configured to hold a heat exchange fluid.

[0210] According to some embodiments of the present disclosure, with reference to FIG. **11**, the battery **100** also includes a second buffering component **70**, and at least a portion of the second buffering component **70** is arranged between the battery cell **20** and the third thermal management component **40**.

[0211] The second buffering component **70** is a component capable of deformation under pressure, and the second buffering component **70** can also be made of an insulating material, thus ensuring electrical insulation between the third thermal management component **40** and the battery cell **20**. This improves the safety of the battery **100**.

[0212] In the embodiment, the second buffering component **70** can be made of rubber or soft polyvinyl chloride.

[0213] The second buffering component **70** can be a sheet-like structure, arranged between the battery cell **20** and the third thermal management component **40**.

[0214] The second buffering component **70** can also wrap around the battery cell **20**, and partially be positioned between the battery cell **20** and the third thermal management component **40**.

[0215] The second buffering component **70** can be arranged on the side of the third thermal management component **40** which is closer to the battery cell **20**, or on the side of the third thermal management component **40** which is farther from the battery cell **20**.

[0216] When the battery cell **20** is provided on both sides of the third thermal management component **40**, the second buffering component **70** can be arranged on both sides of the third thermal management component **40**, or it can only be arranged on one side. The second buffering component **70** can buffer the pressure exerted by the battery cell **20** on the third thermal management component **40** when the battery cell **20** expands, thereby reducing the risk of damage to the third thermal management component **40** due to pressure and enhancing the safety of the battery **100**.

[0217] According to some embodiments of the present disclosure, with reference to FIG. **11**, the second buffering component **70** and the third thermal management component **40** are arranged in a stacked manner.

[0218] The second buffering component **70** can be affixed to the surface wall of the third thermal management component **40** facing the battery cell **20**.

[0219] The second buffering component **70** can abut the third thermal management component **40**, or it can be bonded to the third thermal management component **40**.

[0220] The second buffering component **70** and the third thermal management component **40** are arranged in a stacked manner, which enables the battery cell **20** to preferentially compress the second buffering component **70**, when the battery cell **20** expands in the direction where the third thermal management component **40** is arranged. This reduces the probability of failure or damage to the third thermal management component **40** due to compression. Alternatively, when the third thermal management component **40** is therein arranged with a flow channel for containing heat exchange fluid, the likelihood of the third thermal management component **40** being compressed due to the compression of the flow channel, which further results in poor heat exchange performance, can be reduced.

[0221] According to some embodiments of the present disclosure, referring to FIG. **12** and FIG. **13**, FIG. **12** is a schematic diagram of a cooperation between a second buffering component **70** and a battery cell **20** according to some embodiments of the present disclosure (with a third thermal management component **40** housed in a second housing cavity of a second buffering component **70**, where a first housing cavity is a scaled chamber); and FIG. **13** is a schematic diagram of a cooperation between a second buffering component **70** and a third thermal management component **40** according to some embodiments of the present disclosure (with a second housing cavity of a second buffering component **70** provided with an opening). A second housing cavity is formed inside the second buffering component **70**, and the third thermal management component **40** is housed within the second housing cavity.

[0222] The second housing cavity can be a closed chamber, and the third thermal management component **40** can be completely housed in the second housing cavity.

[0223] The second housing cavity can also be a chamber arranged with an opening, and the third thermal management component **40** can enter the second housing cavity through the opening of the second housing cavity.

[0224] By housing the third thermal management component **40** within the second housing cavity, on the one hand, the risk of disengaging the third thermal management component **40** and the second buffering component **70** is reduced. On the other hand, since the battery cell **20** is directly

against the third thermal management component **40**, the third thermal management component **40** is pressurized and damaged when the battery cell **20** expands; however, this risk is reduced.

[0225] According to some embodiments of the present disclosure, with reference to FIG. **14**, FIG. **14** is a schematic diagram of a cooperation between a third thermal management component **40** and a battery cell **20** according to some embodiments of the present disclosure (with both third surfaces **203** of the battery cell **20** being provided with third thermal management components **40**). The battery cell **20** includes two opposing third surfaces **203**, and the two third surfaces **203** are provided with the third thermal management component **40**.

[0226] The two third surfaces **203** can be in a facing state, meaning the normal of the two third surfaces **203** are collinear. For example, the battery cell **20** can be rectangular in shape, wherein the two third surfaces **203** can be the two surfaces opposite each other in the length direction of the battery cell **20**; and it can also be two surfaces opposite each other in the width direction, and can also be two surfaces opposite each other in the thickness direction.

[0227] The two third surfaces **203** can also be in a non-facing state, where the angle between the normal of the two third surfaces **203** can be less than 90°.

[0228] Each of the two second surfaces **202** can be provided with one third thermal management component **40**, or they can be provided with multiple third thermal management components **40**. By arranging the third thermal management component **40** on both opposing third surfaces **203** of the battery cell **20**, the contact area between the third thermal management component **40** and the battery cell **20** is increased, thereby improving the efficiency of the third thermal management component **40** in adjusting the temperature of the battery cell **20**, and also enhancing the uniformity when regulating the temperature of the battery cell **20**.

[0229] According to some embodiments of the present disclosure, the third surface **203** is the largest surface of the battery cell **20**.

[0230] When the third surface **203** is the largest surface, the contact area between the third thermal management component **40** and the battery cell **20** is maximized, thus improving the efficiency of the third thermal management component **40** in adjusting the temperature of the battery cell **20**.

[0231] According to some embodiments of the present disclosure, with reference to FIG. **2**, the battery **100** includes multiple rows of battery cells **20** and multiple first thermal management components **30**, wherein the multiple first thermal management components **30** are arranged along a first direction. A row of battery cells **20** is arranged between two adjacent first thermal management components **30** along the first direction.

[0232] The first direction can be any one of the length direction, the width direction, or the thickness direction of the battery cell **20**. In the embodiment shown in FIG. **2**, X represents the first direction.

[0233] The two surfaces of the battery cell **20** opposite each other in the first direction can be the two first surfaces **201**.

[0234] In the embodiment, each row of battery cells **20** can include one battery cell **20** or multiple battery cells **20**.

[0235] In the embodiment where each row of the battery cells **20** includes multiple battery cells **20**, the total length of the multiple battery cells **20** along the arrangement direction can be less than or equal to the size of the first thermal management component **30** along the arrangement direction, thus ensuring that each battery cell **20** in each row of the battery cells **20** can fully contact the first thermal management component **30**.

[0236] Since multiple first thermal management components **30** are arranged in the first direction, each row of the battery cells **20** can be housed between two adjacent first thermal management components **30**, thus allowing multiple battery cells **20** to share one first thermal management component **30**. This reduces the number of first thermal management components **30** required for the battery **100**, and saves on the manufacturing cost of the battery **100**. On the other hand, in the embodiment where the battery **100** includes a box **10**, fewer first thermal management components

30 allow for fewer wires or pipelines occupying the internal space of the box **10**. This allows the box **10** to accommodate more battery cells **20**, thereby increasing the energy density of the battery **100**.

[0237] According to some embodiments of the present disclosure, referring to FIG. 2 and FIG. 15, and in conjunction with FIG. 6, FIG. 15 is a schematic diagram of a cooperation between a first buffering component **60** and a battery cell **20** according to further embodiments of the present disclosure (with one first buffering component **60** and one second thermal management component **50** arranged between two battery cells **20**). In each row of battery cells **20**, a second thermal management component **50** is arranged on both sides of each battery cell **20** along a second direction, and a second thermal management component **50** is arranged between two adjacent battery cells **20**. The first direction is perpendicular to the second direction.

[0238] The first direction and the second direction can be any one of the length direction, the width direction, or the thickness direction of the battery cell **20**. In the embodiment shown in FIG. 2, Y represents the second direction.

[0239] The two surfaces of the battery cell **20** opposite each other in the second direction can be the two second surfaces **202**.

[0240] In the embodiment where at least a portion of the first buffering component **60** is arranged between the battery cell **20** and the second thermal management component **50**, between two adjacent battery cells **20**, one second thermal management component **50** can be sandwiched between two first buffering components **60**, or one second thermal management component **50** and one first buffering component **60** can be arranged in a stacked manner.

[0241] By arranging the second thermal management components **50** on both sides of each battery cell **20** in each row of battery cells **20** along the second direction, it ensures that the temperature of each battery cell **20** in each row of battery cells **20** can be regulated by the corresponding second thermal management component **50**. A second thermal management component **50** is arranged between two adjacent battery cells **20** for heating, thus allowing the two battery cells **20** to share one second thermal management component **50**. This reduces the number of second thermal management components **50** required and lowers the manufacturing cost of the battery **100**.

[0242] According to some embodiments of the present disclosure, with reference to FIG. 2, the thermal management component further includes a third thermal management component **40**. The third thermal management component **40** is arranged along the third direction on at least one side of the battery cell **20**, and the third thermal management component **40** is configured to regulate the temperature of the battery cell **20**. The first direction, the second direction, and the third direction are mutually perpendicular.

[0243] In the embodiment shown in FIG. 2, Z represents the third direction.

[0244] The two surfaces of the battery cell **20** opposite each other in the third direction can be the two third surfaces **203**.

[0245] The third thermal management component **40** can correspond one-to-one with each battery cell **20**, or a single third thermal management component **40** can be shared by each row of battery cells **20**, or by all battery cells **20** along the third direction sharing at least one third thermal management component **40**.

[0246] By arranging the third thermal management component **40** on at least one side of the battery cell **20**, it ensures that the temperature of each battery cell **20** in each row of battery cells **20** can be regulated by the corresponding third thermal management component **40**.

[0247] According to some embodiments of the present disclosure, the battery **100** includes at least one first battery cell, at least one second battery cell, a first control module, and a second control module. The thermal management component **90** arranged on the first battery cell is independently controlled by the first control module, and the thermal management component **90** arranged on the second battery cell is independently controlled by the second control module.

[0248] The first battery cell and the second battery cell can be distinguished by their positions. For

example, in an embodiment where the battery **100** includes a box **10**, the box **10** can include multiple peripheral walls connected to the bottom wall, with the first battery cell located between the second battery cell and one of the peripheral walls.

[0249] The first control module and the second control module can be process devices like computers, PLCs, or electrical switches. The working principles of controlling the corresponding thermal management components **90** through the first control module and the second control module are well-known to those skilled in the art and will not be elaborated herein.

[0250] The first control module and the second control module can be two separate devices or two processing circuits within a device.

[0251] This allows for independent temperature regulation of the first battery cell and the second battery cell. Furthermore, different adjustments can be made based on the different environmental temperatures surrounding the first battery cell and the second battery cell, or according to the temperature regulation requirements of each battery cell **20**, thereby improving the rationality of the overall thermal management adjustment of the battery **100**. In the embodiment of the present disclosure, the battery **100** can also include at least one third battery cell, at least one fourth battery cell, and so on, up to at least one N-th battery cell. Correspondingly, the battery **100** can include a third control module, a fourth control module, and so forth up to an N-th control module.

[0252] According to some embodiments of the present disclosure, with reference to FIG. **16**, FIG. **16** is a structural schematic diagram of a battery according to some embodiments of the present disclosure (showing a first battery pack **21** and a second battery pack **22**). The battery **100** includes at least one first battery pack **21**, at least one second battery pack **22**, a third control module, and a fourth control module. The first battery pack **21** includes at least one battery cell **20**, and the second battery pack **22** includes at least one battery cell **20**. The thermal management components **90** arranged on the battery cells **20** of the first battery pack **21** are independently controlled by the third control module, and the thermal management components **90** arranged on the battery cells **20** of the second battery pack **22** are independently controlled by the fourth control module.

[0253] The first battery pack **21** can include a battery cell **20** or multiple battery cells **20**. In this case, the first battery pack **21** can form a module of battery **100** by connecting multiple battery cells **20** in series, in parallel, or in hybrid (which includes both series connection and parallel connection).

[0254] The second battery pack **22** can include a battery cell **20** or multiple battery cells **20**. In this case, the second battery pack **22** can form a module of battery **100** by connecting multiple battery cells **20** in series, in parallel, or in hybrid (which includes both series connection and parallel connection).

[0255] The first battery pack **21** and the second battery pack **22** can be connected in series, in parallel, or in hybrid. The third control module and the fourth control module can be processing devices like computers, PLCs, or electrical switches. The working principles of controlling the corresponding thermal management components **90** through the third control module and the fourth control module are well-known to those skilled in the art and will not be elaborated herein.

[0256] The third control module and the fourth control module can be two separate devices or two processing circuits within a device.

[0257] The first battery pack **21** and the second battery pack **22** can also be multiple battery cells **100** located at different positions within battery **100**. For example, the first battery pack **21** can refer to multiple battery cells **20** near the edge of the box **10** of battery **100**; correspondingly, the second battery pack **22** can refer to multiple battery cells **20** located inside the circle formed by the battery cells **20** of the first battery pack **21**. Alternatively, in an embodiment where the battery cells **20** are housed within a box **10**, the box **10** can be of a rectangular structure, and the first battery pack **21** can refer to the battery cells **20** located in the four corners of the frame of the box **10** of the battery cell **100**; correspondingly, the second battery pack **22** can refer to the battery cells **20** in addition to the four corners. For another example, the first battery pack **21** can refer to the layer of

the battery cells **20** in a multi-layer stacked battery cell **20** near the top cover or bottom wall; correspondingly, the second battery pack **22** can refer to an intermediate layer of the battery cells **20** in the multilayer structure. In the above examples, since the first battery pack **21** is positioned closer to some components of battery **100** that are more prone to heat transfer, the battery cells **20** in this location tend to lose heat more easily than the battery cells **20** in other positions, which causes localized thermal management imbalances. By arranging the thermal management components **90** configured to independently control the battery packs located in different positions within the battery **100**, the thermal management components **90** of the battery pack at the position where the temperature adjustment is needed can be activated independently, and the thermal management components **90** at the battery pack that does not require temperature adjustment can be turned off. This not only conserves thermal management resources but also reduces the thermal impact on battery cells **20** that are already at a stable temperature.

[0258] In the embodiment of the present disclosure, the battery **100** can also include a third battery pack, a fourth battery pack, and so on, up to an N-th battery pack. Correspondingly, the battery **100** can also include a fifth control module, a sixth control module, and so on, up to an N-th control module.

[0259] According to some embodiments of the present disclosure, with reference to FIG. 2, each row of battery cells **20** includes multiple battery cells **20**, and a third thermal management component **40** is arranged to correspond to one row of battery cells **20**, with each third thermal management component **40** being configured to regulate the temperature of one row of battery cells **20**.

[0260] When the third thermal management component **40** is provided in correspondence with the battery cell **20**, the number of the third thermal management component **40** can be the same as the number of rows of the battery cells **20**.

[0261] In the embodiment where each row of the battery cells **20** includes multiple battery cells **20**, the total length of the multiple battery cells **20** along the arrangement direction can be less than or equal to the size of the third thermal management component **40** along the arrangement direction, thus ensuring that each battery cell **20** in each row of the battery cells **20** can fully contact the third thermal management component **40**. For example, assuming that each row of battery cells **20** is arranged in the second direction, the total length of the multiple battery cells **20** along the second direction can be less than or equal to the size of the third thermal management component **40** along the second direction, thus ensuring that each battery cell **20** in each row of the battery cells **20** can fully contact the third thermal management component **40**.

[0262] Each third thermal management component **40** is configured to regulate the temperature of a row of battery cells **20**, which reduces the number of third thermal management components **40** required by the battery **100**. On the one hand, this lowers the manufacturing cost of the battery **100**; and on the other hand, in the embodiment where the battery **100** includes a box **10**, fewer third thermal management components **40** allow for fewer wires or pipelines occupying the internal space of the box **10**. This allows the box **10** to accommodate more battery cells **20**, thereby increasing the energy density of the battery **100**.

[0263] According to some embodiments of the present disclosure, with reference to FIG. 17, FIG. 17 is a schematic diagram of a cooperation between a third thermal management component **40** and a battery cell **20** according to some embodiments of the present disclosure (with a third thermal management component **40** arranged on a side of a battery cell **20** that faces away from an electrode terminal **204** along a third direction). Along the third direction, one side of the battery cell **20** is provided with an electrode terminal **204**, and the side of the battery cell **20** opposite to the electrode terminal **204** is provided with the third thermal management component **40**.

[0264] The electrode terminal **204** is a component of the battery cell **20** for connecting with an external conductor.

[0265] In the embodiment where the battery **100** includes a box **10**, the side on which the electrode

terminal **204** is located can be a side of the battery cell **20** away from the bottom wall.

[0266] By arranging the third thermal management component **40** on the side of the battery cell **20** opposite to the electrode terminal **204**, the third thermal management component **40** can both regulate the temperature of the battery cell **20** and reduce the risk of a third thermal management component **40** forming an interference with the electrode terminal **204**.

[0267] In the embodiment, a fourth thermal management component **80** can be arranged on the side of the battery cell **20** where the electrode terminal **204** is located, and the fourth thermal management component **80** can be arranged between the two electrode terminals **204** of the battery cell **20**.

[0268] According to some embodiments of the present disclosure, the embodiment of the present disclosure further provides an electrical device, comprising the battery **100** provided in the above embodiments.

[0269] According to some embodiments of the present disclosure, with reference to FIG. 2, the embodiment of the present disclosure further provides a battery **100**, which includes a box **10**, multiple battery cells **20**, multiple first thermal management components **30**, multiple second thermal management components **50**, and multiple third thermal management components **40**. The multiple batteries **100** are arranged in multiple rows along a first direction, with the battery cell **20** in each row of the battery cells **20** arranged in a second direction. The multiple first thermal management components **30** are arranged along a first direction. A row of battery cells **20** is arranged between two adjacent first thermal management components **30** along the first direction. The first thermal management component **30** is a water-cooling plate. The multiple second thermal management components **50** are arranged along the second direction on both sides of the battery cell **20**, and the second thermal management components **50** are electric heating components. The box **10** is provided with an opening, and the box **10** is configured to house the battery cells **20**. Along a third direction, the box **10** includes a bottom wall opposite the opening. The third thermal management component **40** is arranged between the battery cell **20** and the bottom wall, with each third thermal management component **40** corresponding to one row of battery cells **20**. The third thermal management components **40** are electric heating components.

[0270] Finally, it should be noted that the above embodiments are only used to illustrate the technical solution of the present disclosure and are not intended to be a limitation thereof. Notwithstanding the detailed description of the present disclosure with reference to the foregoing embodiments, it should be understood by those of ordinary skill in the art that one may still modify the technical solution described in the preceding embodiments, or replace some or all of the technical features equally. These modifications or substitutions do not depart the essence of the corresponding technical solution from the scope of the technical solution of the embodiments of the present disclosure. All of them shall be covered by the scope of the claims and specification of the present disclosure. In particular, as long as there are no structural conflicts, the various technical features mentioned in each embodiment can be combined in any way. The present disclosure is not limited to the specific embodiments disclosed herein but includes all technical solutions falling within the scope of the claims.

Claims

1. A battery, comprising a battery cell comprising a first surface and a second surface; a thermal management component comprising a first thermal management component and a second thermal management component, wherein the first thermal management component is arranged on the first surface, and the first thermal management component is configured to adjust a temperature of the battery cell; and the second thermal management component is arranged on the second surface, and the second thermal management component is configured to adjust the temperature of the battery cell.

2. The battery according to claim 1, wherein the first surface and the second surface are two adjacent surfaces of the battery cell.
3. The battery according to claim 1, wherein: the first thermal management component is configured to heat or cool the battery cell; and/or the second thermal management component is configured to heat the battery cell.
4. The battery according to claim 3, wherein the first thermal management component comprises a first accommodating cavity, and the first accommodating cavity is configured to accommodate a heat exchange fluid.
5. The battery according to claim 3, wherein the second thermal management component is an electric heating component.
6. The battery according to claim 1, further comprising: a first buffering component, wherein at least a portion of the first buffering component is arranged between the battery cell and the second thermal management component.
7. The battery according to claim 6, wherein the first buffering component and the second thermal management component are arranged in a stacked manner.
8. The battery according to claim 6, wherein a first housing cavity is formed inside the first buffering component, and the second thermal management component is housed within the first housing cavity.
9. The battery according to claim 1, wherein the battery cell comprises two opposing first surfaces, and the two first surfaces are provided with the first thermal management component.
10. The battery according to claim 1, wherein the battery cell comprises two opposing second surfaces, and the two second surfaces are provided with the second thermal management component.
11. The battery according to claim 1, wherein the first surface or the second surface is a largest surface of the battery cell.
12. The battery according to claim 1, further comprising: a box with an opening, wherein the box is configured to accommodate the battery cell, the box comprises a bottom wall opposite the opening, and the first thermal management component or the second thermal management component is arranged between the battery cell and the bottom wall.
13. The battery according to claim 1, wherein the battery cell further comprises a third surface, the thermal management component further comprises a third thermal management component, the third thermal management component is arranged on the third surface, and the third thermal management component is configured to adjust the temperature of the battery cell.
14. The battery according to claim 13, wherein the first surface, the second surface, and the third surface are adjacent to each other in pairs.
15. The battery according to claim 13, further comprising: a box with an opening, wherein the box is configured to accommodate the battery cell, the box comprises a bottom wall opposite the opening, and the third thermal management component is arranged between the battery cell and the bottom wall.
16. The battery according to claim 13, wherein the third thermal management component is an electric heating component.
17. The battery according to claim 13, further comprising: a second buffering component, wherein at least a portion of the second buffering component is arranged between the battery cell and the third thermal management component.
18. The battery according to claim 17, wherein the second buffering component and the third thermal management component are arranged in a stacked manner.
19. The battery according to claim 17, wherein a second housing cavity is formed inside the second buffering component, and the third thermal management component is housed within the second housing cavity.
20. An electrical device, comprising the battery according to claim 1.

